THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

RESEARCH / PROJECT PROPOSAL

Campus Resource and Booking Management System

Software Engineering — UCS503

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Submission Date: 18 August 2026

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1. Abstract

Campus rooms and common venues are used throughout the academic year for society preparation, workshops, sessions and events. The proposed Campus Resource and Booking Management System is a web-based system intended to bring the room selection, availability checking, booking request and permission process into one place. The system will cover the campus locations currently identified for booking: LP, LT, TAN, B Block, C Block, D Block, E Block, F Block, Main Auditorium, GR1, GR2 and CR. Rooms and other bookable resources will be maintained under their respective sections.

The proposed workflow starts with a user selecting an existing society, choosing the type of activity, selecting a location and resource, and entering the required date, time and activity details. The user will also select the relevant morning or night permission category. The request will then be reviewed by the designated permission in-charge. Approval will depend on the merits and genuineness of the reason provided in the application. The system will also maintain booking status and prevent conflicting bookings for the same resource and time.

2. Introduction

Booking a campus room is a small task when viewed in isolation, but it becomes more involved when different societies are using the same set of spaces for different activities. A society may need a room for preparation before an event, a workshop may require a particular room for a fixed period, and a larger event may need a venue such as the Main Auditorium. In each case, the request has to be connected to a location, a resource, a date and time, a purpose, and the required permission.

The project proposes a single system for managing this workflow. Rather than treating a booking as only a calendar entry, the system treats it as a request that moves through a defined permission process. This distinction is important because the person responsible for permissions reviews the application before the booking is confirmed.

The project is being developed as part of Software Engineering (UCS503). It therefore focuses not only on building a working interface, but also on requirements engineering, system design, data management, validation, role-based access and testing.

3. Problem Statement

The current project problem is not simply the absence of an online booking form. The larger issue is that a campus resource request contains several connected decisions: which society is requesting the resource, what the activity is, where it will take place, when it will take place, what permission category applies, and whether the request is approved. If these details are handled separately, it becomes harder to get a clear picture of resource availability and request status.

There is also a practical conflict between convenience for the requester and control for the person approving permissions. A requester needs to know whether a suitable resource is available, while the approver needs enough information to judge the request. In the proposed process, the approver will review the application and approve the request when the stated reason is considered genuine and appropriate.

The system therefore needs to connect resource availability with the permission workflow. It must also avoid assigning the same resource to overlapping bookings and should make it clear whether a request is pending, approved, rejected or cancelled.

4. Research Question

How can a centralized campus resource booking system be designed to manage room/resource selection, availability, permission requests and approval in a single workflow for society preparation, workshops/sessions and events?

A secondary question for the later evaluation stage is whether the proposed workflow makes it easier for requesters and the permission in-charge to understand the status of a booking and reduces the possibility of conflicting resource requests. The method and measures for answering this secondary question will be finalized after the project scope and testing plan are agreed.

5. Motivation

The motivation for this project comes from the frequent use of campus spaces by societies, students, and event organisers. Although the booking process is similar for different activities, the information and requirements may vary. A simple booking may need only basic details, while events and workshops may require additional information about participants, purpose, and resources.

A centralized system can make the process more organised and transparent by allowing requesters to view available resources and track applications, while the permission in-charge can review and manage requests in one place. The project also provides a practical application of software engineering concepts to a real campus problem.

6. Objectives

The project will aim to:

- Develop a centralized system for requesting and managing campus rooms and other bookable resources.
- Represent the campus locations and their rooms/resources in a structured hierarchy.
- Allow a user to select an existing society from a maintained list.
- Support the three identified activity categories: society preparation, workshop/session and event.
- Allow users to select a location, resource, date and time before submitting a request.
- Support morning and night permission categories, with the exact timing rules to be finalized.
- Provide a clear approval workflow for the designated permission in-charge.
- Record the reason and other relevant details of each booking request so that the approver can review it.
- Prevent conflicting bookings for the same resource and overlapping time period.
- Allow users to track the status and history of their requests.
- Provide administrative controls for maintaining locations, resources, societies and relevant system data.

7. Scope of the Proposed System

- Campus resource and room booking.
- Location → resource hierarchy covering LP, LT, TAN, B Block, C Block, D Block, E Block, F Block, Main Auditorium, GR1, GR2 and CR.
- Booking for society preparation, workshops/sessions and events.
- Selection of an existing society.
- Date and time selection.
- Morning/night permission category.
- Permission review and approval/rejection.

- Availability and conflict checking.
- Booking status and booking history.
- Administrative maintenance of resources and related records.

8. Initial Functional Requirements

ID	Requirement	Description
FR-01	User and society access	The system shall allow an authorized user to access the booking workflow and select an existing society from the available society list.
FR-02	Activity selection	The system shall allow the user to identify the request as society preparation, workshop/session or event.
FR-03	Location selection	The system shall display the configured campus booking locations.
FR-04	Resource selection	The system shall display the rooms/resources associated with the selected location.
FR-05	Date and time	The system shall collect the requested date, start time and end time.
FR-06	Permission category	The system shall allow the requester to select morning or night permission.
FR-07	Activity details	The system shall collect the information needed for the selected activity type.
FR-08	Availability check	The system shall check whether the requested resource has a conflicting booking for the selected time.
FR-09	Request submission	The system shall create a booking request with a trackable status.
FR-10	Approval	The system shall provide the designated permission in-charge with the ability to review and approve or reject requests.
FR-11	Reason for decision	The system shall support recording a reason or comment when a request is rejected or when additional explanation is required.
FR-12	Status tracking	The system shall allow users to see the current status of their requests.
FR-13	Cancellation	The system shall support cancellation of a booking/request, subject to the final rules decided during requirements analysis.
FR-14	Administration	The system shall provide administrative management of locations, rooms/resources, societies and relevant records.

9. Proposed Methodology

The project will follow an iterative software development approach. This is a proposed project decision rather than a statement about the current campus process. The work will be divided into requirements, design, implementation, testing and refinement. Requirements will be documented before the corresponding feature is treated as complete, and feedback from testing will be used to refine the next iteration.

9.1 Requirements analysis

The team will identify the booking types, locations, resources, permission rules and information required for each request. The open items identified in this proposal will be confirmed before the final requirements baseline is prepared. Requirements will be expressed in a form that can be traced to design and testing.

9.2 System design

The team will prepare the system architecture, use-case model, activity flow, database/entity model and interface structure. Particular attention will be given to the relationship between locations, resources, bookings and permission decisions.

9.3 Implementation

The system will be implemented as a web application using React.js for the frontend, Node.js with Express.js for the backend, and PostgreSQL for database management. JWT will be used for authentication and security, while Git and GitHub will support version control and collaboration.

9.4 Validation and testing

Testing will cover the booking workflow, access control, input validation, resource availability, overlapping time periods, approval/rejection states, cancellation and administrative operations.

9.5 Iteration

Findings from testing will be used to correct the system and refine requirements where necessary. This is important for a booking system because errors in time handling or status transitions can affect the reliability of the entire workflow.

10. Proposed System Workflow

The initial workflow is:

Login → Select Society → Select Activity Type → Select Location → Select Resource → Select Date & Time → Select Morning/Night Permission → Enter Activity Details → Check Availability → Submit Request → Permission In-charge Review → Approve/Reject → Confirmed / Rejected Booking

The approval step is intentionally separate from submission. The requester can submit the application, but the booking becomes confirmed only after the designated permission in-charge reviews the request. The project team has specified that the approver will consider the genuineness of the stated reason when deciding whether to approve a request.

11. Initial Data Model

The following entities are proposed as the starting point for database design. The exact fields will be finalized during requirements and design.

Entity	Purpose
User	Stores user identity, login information and role.
Society	Stores the societies available for selection.
Location	Stores campus sections such as LP, LT, TAN and the listed blocks/venues.
Resource	Stores rooms and other bookable resources belonging to a location.

Booking	Stores the requested resource, date, time, activity type, purpose and current status.
Permission	Stores the approval/rejection decision, approver and decision details.
Maintenance / Unavailability	Stores periods when a resource cannot be booked, if this feature is included.

12. User Roles

Role	Main responsibility
Requester / User	Select society, choose activity and resource, submit requests, view status and cancel requests where permitted.
Permission In-charge	Review submitted applications and approve or reject them based on the information and reason provided.
Administrator	Maintain system data such as societies, locations, rooms/resources and user access.

13. Expected Outcomes

The expected outcome is a working prototype that represents the campus booking process as one connected workflow. A user should be able to move from society and activity selection to a resource request and then track the permission decision. The system should also maintain a reliable record of resource usage and prevent confirmed overlapping bookings.

- A structured catalogue of campus locations and bookable resources.
- A booking workflow tailored to preparation, workshop/session and event requests.
- A defined permission and approval workflow.
- Availability and overlap validation.
- User-facing booking status and history.
- Administrative management of the core resource data.
- Software engineering artefacts such as requirements, system design and test cases.

14. Risks and Mitigation

Risk	Mitigation
Incomplete or changing requirements	Confirm the open operational rules before freezing the requirements baseline and keep changes traceable.
Incorrect resource inventory	Use the verified campus resource list and treat the current location list separately from the room inventory still to be confirmed.
Ambiguity in permission timings	Keep morning/night as configurable categories until exact timings are confirmed.
Booking conflicts	Perform overlap validation on the server-side booking logic, not only in the user interface.
Unclear approval rules	Document the approver's decision criteria during requirements analysis rather than inventing additional rules.
Scope growth	Keep optional features separate from the core booking and permission workflow until the core system is stable.